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ENGINEERING
COLLEGE OF ENGINEERING
SEOUL NATIONAL UNIVERSITY
서 울 대 학 교 공 과 대 학

JEONG WOO HAN, PH.D.

EDUCATION

Jan 2008 – Mar 2010 **Georgia Institute of Technology** Atlanta, GA
Doctor of Philosophy in Chemical and Biomolecular Engineering
Advisor: Prof. David S. Sholl
Thesis: *Density Functional Theory Studies for Separation of Enantiomers of a Chiral Species by Enantiospecific Adsorption on Solid Surfaces*

Aug 2006 – Dec 2007 **Carnegie Mellon University** Pittsburgh, PA
Master of Science in Chemical Engineering
Advisor: Prof. David S. Sholl
Thesis: *First Principles Studies of Molecular Adsorption on Hydroxylated Quartz(0001)*

Mar 1998 – Aug 2005 **Seoul National University** Seoul, Korea
Bachelor of Science in Chemical and Biological Engineering

EXPERIENCE

Seoul National University Seoul, Korea
Sep 2023 – Present Associate Professor of Materials Science and Engineering
Mar 2024 – Present Vice Head of Research Affairs / BK21

Pohang University of Science and Technology (POSTECH) Pohang, Korea
Aug 2022 – Aug 2023 Mueunjae Chair Professor
Mar 2022 – Aug 2023 Professor of Chemical Engineering / Young Distinguished Professor
Feb 2018 – Feb 2022 Associate Professor of Chemical Engineering
July 2018 – Aug 2023 Adjunct Professor of Interdisciplinary Bioscience and Bioengineering
Mar 2020 – Aug 2023 Adjunct Professor of Artificial Intelligence
Sep 2021 – Aug 2023 Adjunct Professor of Environmental Science and Engineering

Jan 2022 – Jan 2023 **SLAC National Accelerator Laboratory & Stanford University** Menlo Park, CA
Visiting Professor of SUNCAT Center for Interface Science and Catalysis

Yonsei University Seoul, Korea
Oct 2018 – Aug 2023 Adjunct Professor of Convergence Research and Education in Advanced Technology

Jan 2020 – Dec 2024 **Member of Young Korean Academy of Science and Technology (Y-KAST)**

University of Seoul Seoul, Korea
Sep 2016 – Jan 2018 Associate Professor of Chemical Engineering
Sep 2012 – Aug 2016 Assistant Professor of Chemical Engineering
Apr 2016 – Jan 2018 Director, Division of Chemistry
Jun 2016 – Feb 2017 Director, Center for International Cooperation

Dec 2022 – Present **Advisory Board Member, *Carbon Energy*** Wiley

Feb 2017 – Dec 2024 **Editor, *Molecular Catalysis*** Elsevier, B.V.

Jan 2021 – Present **Associate Editor, *Korean Journal of Chemical Engineering*** Springer

Apr 2010 – July 2012 **Massachusetts Institute of Technology** Cambridge, MA
Department of Nuclear Science and Engineering
Postdoctoral Associate working with Prof. Bilge Yildiz

Aug 2005 – Jul 2006 **Korea Institute of Science and Technology (KIST)** Seoul, Korea
Commissioned Research Scientist working with Dr. Soo Hyun Kim

RECENT SELECTED PUBLICATIONS (*Last 5 years; Last corresponding authored papers only; Most recent IF*)

1. Computationally guided dopant discovery for boosting TiO_2 catalysts in plasma-assisted non-oxidative coupling of methane
   Hyeonae Im†, Hoang Phuong Nguyen†, Yunkyung Kim†, Juchan Kim, Chaesung Lim, Yongju Yun, Kyoung-Su Ha*, and **Jeong Woo Han***
   Appl. Catal. B-Environ. Energy **383** (2026) 126051. (IF=21.1)
2. Strategic Design of Oxophilic Dopants for Active and Durable Alkaline Hydrogen Evolution Reaction under Seawater
   Hyeonjung Jung†, Yechan Lee†, Kyuri Cho†, Tae Yong Kim, Jihyeon Song, Han Sol Jung, Sung Il Park, Yoojin Lee, Jinuk Moon, Wongyu Park, Jaewoo Nam, Sangmin Park, Wooyul Kim, and **Jeong Woo Han***
   Adv. Funct. Mater. **in press** (2025). (IF=19.0)
3. Synergistic Fe,Pd diatomic sites anchored on porous nitrogen-doped carbon for efficient oxygen reduction in whole pH range
   Yejung Choi†, Byoung Joon Park†, Yechan Lee, Kug-Seung Lee*, Jin Kon Kim*, and **Jeong Woo Han***
   J. Mater. Chem. A **13** (2025) 21462–21471. (IF=9.5)
   *Selected as an *Inside front cover*
4. Rational design of nitrogen-doped porous carbon support on single atom catalysts for efficient CO_2 electroreduction

Yejung Choi†, Keon-Woo Kim†, Byoung Joon Park, Tae Yong Kim, Yechan Lee, Bomi Park, Yechan Lee, Jin Kon Kim*, and **Jeong Woo Han***
J. Mater. Chem. A **13** (2025) 4861–4869. (IF=9.5)
*Selected as a *Front cover*

5. Boosting the catalytic activity toward oxygen reduction via a heterostructure of porous iron oxide-decorated 2D NiO/NG nanosheets
Kakali Maiti, Matthew T. Curnan, Hyung Jun Kim, Kyeounghak Kim*, and **Jeong Woo Han***
J. Energy Chem. **93** (2024) 669–681. (IF=14.9)
6. Tailoring local structures of atomically dispersed copper sites for highly selective CO_2 electroreduction
Kyung-Jong Noh†, Byoung Joon Park†, Ying Wang, Yejung Choi, Sang-Hoon You, Yong-Tae Kim, Kug-Seung Lee*, and **Jeong Woo Han***
Carbon Energy **6** (2024) e419. (IF=24.2)
7. Computational Discovery of Optimal Dopants for Nickel Iron Oxyhydroxide to Enhance OER Activity and Saline Water Compatibility
Hyeonjung Jung†, Jihyeon Song†, Yechan Lee, Han Sol Jung, Kyung-Jong Noh, Hyeonae Im, Yoojin Lee, Tae Yong Kim, Okkyun Seo, Takeshi Watanabe, Loku Singgappulige Rosantha Kumara, Daiju Matsumura, Sangmin Park, and **Jeong Woo Han***
ACS Energy Lett. **9** (2024) 2162–2172. (IF=18.2)
*Selected as a *Supplementary cover*
8. Roles of bridge carbon and heteroatom during the dehydrogenation reaction of dicyclohexylmethane derivatives on Pd and Pt catalysts for liquid organic hydrogen carrier
Hyunwoo Yook†, Kwanyong Jeong†, Jinwoo Hwang, Ji Hoon Park*, and **Jeong Woo Han***
Appl. Catal. B-Environ. **342** (2024) 123394. (IF=22.1)
9. Accelerating Li-based battery design by computationally engineering materials
Sandip Maiti†, Matthew T. Curnan†, Kakali Maiti†, Seokhyun Choung, and **Jeong Woo Han***
Chem **9** (2023) 3415–3460. (IF=19.6)
10. Design Strategies for Hydroxyapatite-based Materials to Enhance their Catalytic Performance and Applicability
Hyunwoo Yook†, Jinwoo Hwang†, Woonsuk Yeo†, Jungup Bang, Jaeyoung Kim, Tae Yong Kim*, Jae-Soon Choi*, and **Jeong Woo Han***
Adv. Mater. **35** (2023) 2204938. (IF=26.8)
11. Tuning Local Coordination Environments of Manganese Single-Atom Nanozymes with Multi-Enzyme Properties for Selective Colorimetric Biosensing
Ying Wang†, Ara Cho†, Guangri Jia, Xiaoqiang Cui, Junhyeop Shin, Inho Nam, Kyung-Jong Noh, Byoung Joon Park, Rui Huang, and **Jeong Woo Han***
Angew. Chem. Int. Ed. **62** (2023) e202300119. (IF=16.9)

12. Change in the Electronic Environment of the VO_x Active Center via Support Modification to Enhance Hg Oxidation Activity
    Woonsuk Yeo†, Dongjae Shin†, Moon Hyeon Kim, and **Jeong Woo Han***
    ACS Catal. **13** (2023) 3775–3787. (IF=13.1)
    *Selected as a *Supplementary cover*
13. Dynamic Surface Evolution of Metal Oxides for Autonomous Adaptation to Catalytic Reaction Environments
    Jun Kyu Kim†, Sangwoo Kim†, Seunghyun Kim, Hyung Jun Kim, Kyeounghak Kim*, WooChul Jung*, and **Jeong Woo Han***
    Adv. Mater. **35** (2023) 2203370. (IF=26.8)
    *Selected as a *Frontispiece*
14. Modulating the water gas shift reaction via strong interfacial interaction between a defective oxide matrix and exsolved metal nanoparticles
    Huijun Chen†, Rui Huang†, Myeong Gon Jang†, Chaesung Lim, Qiuyu Liu, Heejae Yang, Yan Chen*, and **Jeong Woo Han***
    J. Mater. Chem. A **10** (2022) 24995–25008. (IF=9.5)
    *Selected as a *Back cover*
15. Role of an Interface for Hydrogen Production Reaction over Size-Controlled Supported Metal Catalysts
    Dongjae Shin†, Rui Huang†, Seokhyun Choung, Youngbi Kim, Kiheon Sung, and **Jeong Woo Han***
    ACS Catal. **12** (2022) 8082–8093. (IF=13.1)
16. Boosting Support Reducibility and Metal Dispersion by Exposed Surface Atom Control for Highly Active Supported Metal Catalysts
    Myeong Gon Jang†, Sinmyung Yoon†, Dongjae Shin†, Hyung Jun Kim, Rui Huang, Euiseob Yang, Jihun Kim, Kug-Seung Lee, Kwangjin An*, and **Jeong Woo Han***
    ACS Catal. **12** (2022) 4402–4414. (IF=13.1)
17. Precisely Constructing Orbital Coupling-Modulated Dual-Atom Fe Pair Sites for Synergistic CO_2 Electroreduction
    Ying Wang†, Byoung Joon Park†, Vinod K. Paidi, Rui Huang, Yechan Lee, Kyung-Jong Noh, Kug-Seung Lee*, and **Jeong Woo Han***
    ACS Energy Lett. **7** (2022) 640–649. (IF=18.2)
    *Selected as a *Supplementary cover*
18. Systematic Approach to Designing a Highly Efficient Core-Shell Electrocatalyst for N_2O Reduction
    Kyeounghak Kim†, Jinuk Byun†, Hyunjoong Kim†, Kug-Seung Lee, Hyeon Seok Lee, Jiheon Kim, Taeghwan Hyeon*, Jae Jeong Kim*, and **Jeong Woo Han***
    ACS Catal. **11** (2021) 15089–15097. (IF=13.1)
    *Selected as a *Front cover*
19. Effective Screening Route for Highly Active and Selective Metal–Nitrogen–Doped Carbon Catalysts in CO_2 Electrochemical Reduction

Byoung Joon Park†, Ying Wang†, Yechan Lee†, Kyung-Jong Noh, Ara Cho, Myeong Gon Jang, Rui Huang, Kug-Seung Lee, and **Jeong Woo Han***
Small **17** (2021) 2103705. (IF=12.1)
*Selected as a *Front cover*

20. Engineering Single Atom Catalysts to Tune Properties for Electrochemical Reduction and Evolution Reactions
Kakali Maiti†, Sandip Maiti†, Matthew T. Curnan, Hyung Jun Kim, and **Jeong Woo Han***
Adv. Energy Mater. **11** (2021) 2101670. (IF=26.0)
*Selected as a *Front cover*
21. Engineering electrocatalyst nanosurfaces to enrich activity by inducing lattice strain
Sandip Maiti†, Kakali Maiti†, Matthew T. Curnan, Kyeounghak Kim, Kyung-Jong Noh, and **Jeong Woo Han***
Energy Environ. Sci. **14** (2021) 3717–3756. (IF=30.8)
*Selected as an *Insider front cover*
22. Synergistic coupling ensuing cobalt phosphosulfide encapsulated by heteroatom-doped two-dimensional graphene shell as an excellent catalyst for oxygen electroreduction
Kakali Maiti, Kyeounghak Kim, Kyung-Jong Noh, and **Jeong Woo Han***
Chem. Eng. J. **423** (2021) 130233. (IF=13.2)
23. Mechanistic insights into phase transition and metal ex-solution phenomena of $Pr_{0.5}Ba_{0.5}Mn_{0.85}Co_{0.15}O_{3-\delta}$ from simple to layered perovskite under reducing conditions and enhanced catalytic activity
Kyeounghak Kim, Sangwook Joo, Rui Huang, Hyung Jun Kim, Guntae Kim*, and **Jeong Woo Han***
Energy Environ. Sci. **14** (2021) 873–882. (IF=30.8)
*Selected as a *Back cover*

FULL PUBLICATION LIST (*Published year's IF*)

1. A novel one-step chemo-bioconversion to produce 1,3-butanediol from polyethylene terephthalate (PET)
Ji-yeun Yi†, Mi Hee Ryu†, Youngbi Kim†, Yeonkyeong Ryu†, Eunchae Song, Jung-Hoon Bae, Jung-Hoon Sohn, **Jeong Woo Han***, Hyun Gil Cha*, Hee Taek Kim*, and Bong Hyun Sung*
submitted (2025).
2. Cu/Cu_2O interface-modulated two-stage reduction of Cu_2O under methanol
Meng Li, Matthew T. Curnan, Hao Chi, Stephen D. House, Christopher M. Andolina, **Jeong Woo Han**, Wissam A. Saidi, Götz Veser, and Judith C. Yang
submitted (2025).
3. Aqueous-phase Methane Oxidation with H_2O_2 over Fe-zeolites
Min Sik Kim, Woonsuk Yeo, Chang Shuai, Seok Young Hwang, Tae Yong Kim, Sung June Cho*, **Jeong Woo Han***, and Eun Duck Park*

submitted (2025).

4. Visualization of Cluster-mediated Oxidation on Cu(111)
   Hao Chen†, Matthew T. Curnan†, Qingfei Liu†, Yunjian Ling, Jiayu Lv, Yangsheng Li, Wugen Huang, Changbao Zhao, Rui Wang, Zhongmiao Gong, Yi Cui, **Jeong Woo Han**, Judith C. Yang, Wissam A. Saidi, Miquel Salmeron, and Fan Yang
   submitted (2025).
5. Engineered Metal-Metal Interfaces from Layered Double Hydroxide Frameworks Enhance Solid Oxide Cell Performance
   Hyunmin Kim†, Zhengyu Wu†, Yoon Seo Kim†, Hwakyoung Seo†, Minhao Zhang, Hyong June Kim, Chaesung Lim, **Jeong Woo Han***, Seungho Cho*, Jihwan An*, and Yunfei Bu*
   submitted (2025).
6. Phase-transformation-enabled linear junctions drive high-rate electrosynthesis of H_2O_2
   Hyunmin Kim†, Seokhyun Choung†, Rong Ma†, Zhirui Fu, **Jeong Woo Han***, Jihwan An*, and Yunfei Bu*
   submitted (2025).
7. Sustainable and high-performance polyurethane foams produced using waste-derived polyols
   Harin Kim, Jiyoung Chae, Yoonjeong Jo, Su Min Jung, Hyeokjoon June, Jun Dong Park, **Jeong Woo Han***, Kwangjin An*, and Joon Hyun Baik*
   submitted (2025).
8. Lattice Distortion in Perovskite Oxides Boosts the Metal Exsolution
   Yo Han Kim†, Uchan Jeon†, Hyeongwon Jeong, Bo-Ram Won, **Jeong Woo Han***, and Jae-ha Myung*
   submitted (2025).
9. Lattice-Confined Ruthenium Nanoparticles for Reverse-Current Tolerant Hydrogen Evolution in Alkaline Electrolyzers
   Sang-Mun Jung†, Kyu-Su Kim†, Jinwoo Baek, Yoojin Lee, In-Ho Baek, Byung-Jo Lee, Gunjoo Kim, Gogwon Choe, Jinhyeon Lee, Jaerim Kim, Sang-Hoon You, Haeyong Shin, Young-Woo Kim, Kug-Seung Lee, Hyunjoo Lee, **Jeong Woo Han**, and Yong-Tae Kim
   submitted (2025).
10. Electronic Activation at Heterostructures for Efficient High Temperature CO_2 Electrolysis
    Mengzhen Zhou†, Benchi Chen†, Chaesung Lim†, Xiang Sun, Xuepeng Xiang, Yongjian Ye, Hui Zhang, Nian Zhang, **Jeong Woo Han***, and Yan Chen*,
    submitted (2025).
11. Quantitative Electrochemical Evaluation of Metal Nanocatalysts for Oxygen Exchange on Solid Oxide Cell Electrodes
    Jinwook Kim†, Geon Gug Yang†, WonJun Lee†, Seongwoo Nam, Heejae Yang, Jongsu Seo, Beomgyun Jeong*, **Jeong Woo Han***, Sang Ouk Kim*, and WooChul Jung*
    submitted (2025).

12. Unlocking Exceptional Antibacterial Potency in Root Canal Sealers via $AgCuO_2$ Nanocomposites
    Seyeon Kim†, Min–Yong Lee†, Hwakyoung Seo†, Tae Yong Kim, Kanghyun Lee, Yujin Lee, Abraham Seo, Eo–Jin Kang, Ji–Yeon Hong, Inho Nam, Jungu Kang, Seung Hyun Cho, **Jeong Woo Han***, Jae–Sung Kwon*, and Soomin Park*
    submitted (2025).
13. Gas-Phase Engineered W Single-Atom Catalysts: Pathways to Enhanced Catalytic Performance for Oxidative Coupling of Methane
    Heeyeon Kim*,†, Hyeonae Im†, **Jeong Woo Han***, Seoyoung Jang, Min Seon Choi, Akhil Tayal, and Okkyun Seo
    submitted (2025).
14. Highly Selective Electrochemical Reduction of CO_2 to Ethanol Enabled by a Mn–Mn Single-Atom Dimer Catalyst
    Taehun Yang, Silambarasan Perumal, Miyeon Kim, Eunji Lee, Kainat Talat, Umair Muhammad, Seungjune Lee, Sehyun Joung, Ta Thi Thuy Nga, Mansoor Ali, Chung-Li Dong, **Jeong Woo Han**, and Hyoyoung Lee*
    submitted (2025).
15. Chelation-enabled efficient CO_2 electroreduction to methane in natural seawater using low-coordinated Cu catalyst
    Donggeun Eom†, Yoojin Lee†, **Jeong Woo Han***, and Sangwook Park*
    Nat. Commun. submitted (2025).
16. Tailoring Coordination Environments of Rh/TiO_2 Single-Atom Catalysts via Cation Exchange for Superior Hydroformylation Performance beyond Homogeneous Catalysts
    Daewon Oh†, Wonsik Jang†, Miyeon Kim†, Hojeong Lee, Jin Ho Lee, Beomgyun Jeong, Jeongjin Kim, Jueun Kim, **Jeong Woo Han***, Seungho Cho*, and Kwangjin An*
    Angew. Chem. Int. Ed. submitted (2025).
17. Niobium-modified nickel electrocatalysts with enhanced reverse-current tolerance for intermittent alkaline water electrolysis
    Sang-Mun Jung†, Byung-Jo Lee†, Yoojin Lee†, Wonjoon Choi, Kyu-Su Kim, Jaeik Kwak, Hoyeon Shin, Jinwoo Baek, In-Ho Baek, Haeyong Shin, Sang-Hoon You, Hyun-Yup Kim, **Jeong Woo Han***, and Yong-Tae Kim*
    Joule submitted (2025).
18. Enabling photocurrent density over 8.1 mA cm^{-2} using iron oxide with in situ glycerol valorization
    Sarang Kim†, Juhyung Park†, Jinwoo Hwang†, **Jeong Woo Han***, Ji-Hyun Jang*, and Ji-Wook Jang*
    Nat. Commun. submitted (2025).
19. Li_3PO_4-assisted Direct Recycling of NMC Cathodes: Performance Recovery of Spent Li-ion Batteries
    Jongkwon Park, Seyoung Choi, Junhyeop Shin, Jae Hyun Kim, Myeong-Lok Seol, Hannah Song, **Jeong Woo Han**, Simon MoonGeun Jung, Pyung Soo Lee, and Inho Nam

Adv. Energy Mater. submitted (2025).

20. Diverse Cation Exchange in Layered Titanate Nanostructures for Tailored Catalysis
    Hyoseok Kim†, Daewon Oh†, Miyeon Kim†, Taein Lee, Sihyeon Baek, Wonsik Jang, Jong-Seong Bae, Jihun Kim, Jongkyoung Kim, Taeheon Kim, Changhun Hur, Sinmyung Yoon, Hu Young Jeong*, **Jeong Woo Han***, Kwangjin An*, and Seungho Cho*
    Adv. Mater. submitted (2025).
21. Intrinsic reversibility in Ir-based electrocatalysts via dynamic segregated-surface reconstruction
    Sang-Hoon You†, Hyeonjung Jung†, Kyu-Su Kim, Jinwoo Baek, Gogwon Choe, Jaekyung Yi, Ki-Jeong Kim, Sarah S. Park, **Jeong Woo Han***, Kug-Seung Lee*, and Yong-Tae Kim*
    Energy Environ. Sci. submitted (2025).
22. Molten GaBi Catalyst for Sustainable Hydrogen Production at Moderate Temperatures by Methane Pyrolysis
    Junseok Song†, Hosik Hahn†, Sung-Hoon Gwak, Young-Woon Byeon, **Jeong Woo Han***, and Sangwook Park*
    Chem. Eng. J. submitted (2025).
23. Dual-Amide Strategy for Quaternary Aqueous Electrolytes
    Insu Jeong†, Younsang Cho†, Youngbi Kim†, **Jeong Woo Han***, Soojin Park*, and Jaegeon Ryu*,
    Adv. Energy Mater. submitted (2025).
24. Spin-State Modulation via Asymmetric Ni Single-Atom Sites from bio-MOF-1 for Highly Selective CO_2 Electroreduction
    Yechan Lee†, Byoung Joon Park†, Sang-Hoon You†, Changshin Jo, Kug-Seung Lee, Yong Tae Kim, and **Jeong Woo Han***
    Nano Energy submitted (2025).
25. Nanoporous IrFe Thin-Film Catalyst for Water Electrolysis with a Longevity of Two Decades
    Kyu-Su Kim†, Hyeonjung Jung†, Kyeong-Rim Yeo†, Sang-Mun Jung†, Sang-Hoon You, Jinhyeon Lee, Jaekyung Yi, Seung Woo Lee, Chanho Park, Sarah S. Park, Ki-Jeong Kim, Kug-Seung Lee, Docheon Ahn, Soo-Kil Kim*, **Jeong Woo Han***, and Yong-Tae Kim*
    Science submitted (2025).
26. Hierarchical ceria nanoarchitecture enabling accelerated lattice oxygen dynamics for advanced redox reactions
    Seokhyun Choung†, Myeong Gon Jang†, Yunkyung Kim, Taein Lee, Wongyu Park, Miyeon Kim, Okkyun Seo, Takeshi Watanabe, Loku Singgappulige Rosantha Kumara, Daiju Matsumura, Tae Yong Kim, and **Jeong Woo Han***
    Nat. Catal. **in revision** (2025). (IF=44.6)
27. Light oils from methane homologation over a hierarchically porous titania catalyst for a non-thermal plasma
    Minkyeong Ban†, Juchan Kim†, Hyeonae Im†, Seongseop Kim, Yongju Yun, **Jeong Woo Han***, Kyoung-Su Ha*, and Jinwoo Lee*

Appl. Catal. B-Environ. Energy **in revision** (2025). (IF=21.1)

28. From Atomic Motif to Realistic Single Atom Catalysts through Machine Learning Interatomic Potentials
    Seokhyun Choung†, Miyeon Kim†, Jinuk Moon†, and **Jeong Woo Han***
    ACS Energy Lett. **in revision** (2025). (IF=18.9)
29. Dissociative oxygen reduction on FeCo dual-atom catalysts for enhanced Zn–air battery performance
    Byoung Joon Park†, Yejung Choi†, Kyung-Jong Noh†, Junhyeop Shin, Kyuri Cho, Wooyul Kim*, Jin Kon Kim*, Inho Nam*, and **Jeong Woo Han***
    Appl. Catal. B-Environ. Energy **in revision** (2025). (IF=21.1)
30. *OH adsorption-mediated electrochemical oxidation of 5-hydroxymethylfurfural to selective 2,5-furandicarboxylic acid in weakly alkaline media
    Eunchong Lee†, Jinwoo Hwang†, Suhwan Yoo, Juhyung Choi, Hyun Ji An, Sang Heon Han, Taerin Kim, **Jeong Woo Han***, and Yun Jeong Hwang*
    Adv. Sci. **in revision** (2025). (IF=14.1)
31. Doping-enabled Symmetry Control in $BaCoO_3$ for Enhanced Oxygen Reduction Reaction
    Sejong Ahn†, Kyu In Shim†, Hyunseung Kim, Seongwoo Nam, Yeongtaek Hong, DongHwan Oh, Jun Hyuk Kim*, **Jeong Woo Han***, and WooChul Jung*
    Small **in revision** (2025). (IF=12.1)
32. Enhanced Kinetics and Dual-Interfacial Stability in High-Areal-Loading Electrodes for Lithium Metal Batteries via Crosstalk-Active Polymer-Functionalized Carbon Nanotubes
    Jaeho Jung†, Sungho Kim†, Youngbi Kim†, Seoha Nam, Yeongseok Kim, Jeongyun Bae, **Jeong Woo Han***, and Soojin Park*
    Adv. Mater **in revision** (2025). (IF=26.8)
33. Decoding the full catalytic cycle of class I PHA synthase: A QM/MM metadynamics perspective
    Youngbi Kim, Hyomin Lee, Hyewon Lee, and **Jeong Woo Han***
    Int. J. Biol. Macromol. **in revision** (2025). (IF=8.5)
34. Overcoming the Lattice Mismatch Barrier for Atomic Reconstruction in $MoSe_2/MoS_2$ Heterobilayers
    Seong Chul Hong†, Ji-Hwan Baek†, Yunyeong Chang†, Manh Hong Nguyen, Wongyu Park, Chaesung Lim, Jinwoo Kim, Hyeongseok Lee, Changheon Kim, Kenji Watanabe, Takashi Taniguchi, Jangyup Son, **Jeong Woo Han**, Hyeonsik Cheong, Miyoung Kim, and Gwan-Hyoung Lee
    ACS Nano **in revision** (2025). (IF=16.0)
35. TiO_2-facet-dependent reconstruction of Pt nanoparticles during CO oxidation
    Jihun Kim†, Yunkyung Kim†, Daewon Oh, Seokhyun Choung, Hyeongeon Lee, Hogyun Lee, Eonu Nam, Shinjae Lee, Jeongjin Kim, Sinmyung Yoon*, **Jeong Woo Han***, and Kwangjin An*
    Nat. Commun. **in revision** (2025). (IF=15.7)

36. Electrodeposition-driven metal-support coupling in Pd-CeO_2 on SiC monolith for low power joule-heated CO oxidation
    Hyunji Lee, Jina Kim, Sohyun Kang, Kyung-Min Kim, Ju Hwan Kim, Dong Gwon Kang, **Jeong Woo Han**, Jungmin Lee, Young-Woo You, Young Jin Kim, Hyun-Tak Kim, and Sang-Joon Kim
    Chem. Eng. J. **in revision** (2025). (IF=13.2)
37. Comprehensive Understanding of Accelerated Kinetics Driven by Anion–Diluent Dynamics Enabling Wide Temperature Operation in Dual-Ion Batteries
    Sungho Kim†, Youngbi Kim†, Heejae Yang, Dongjoo Kim, Jieun Kang, Dokyung Kim, Young Joo Lee, **Jeong Woo Han***, Jaegeon Ryu*, and Soojin Park
    Adv. Mater. **in revision** (2025). (IF=26.8)
38. Breakthrough oxygen electrode reaction kinetics using $BaCe_{0.7}Zr_{0.1}Y_{0.1}Yb_{0.1}O_{3-\delta}$ based composite air electrodes for reversible solid oxide cells
    Kwangho Park†, WonJun Lee†, Junghyun Park, Minkyeong Jo, Yeongeun Bae, Minji Kim, Ryan O'Hayre*, **Jeong Woo Han***, Sun-ju Song*, and Jun-Young Park*
    Nat. Energy **in revision** (2025). (IF=60.1)
39. Bridging Polymer Electrolyte Design and Electrode Architecture for Stable Operation of Polymer-based Solid-State Lithium Metal Batteries
    Yeong Hun Choi†, Sungho Kim†, Hyojun Lee†, Youngbi Kim, Seoha Nam, Dong-Yeob Han, Jaeho Jung, Yeongseok Kim, **Jeong Woo Han***, and Soojin Park*
    Adv. Mater. **in revision** (2025). (IF=26.8)
40. Breaking the optical loss barrier in amorphous silicon across the full visible spectrum via dopant-controlled chemical vapor deposition
    Younghwan Yang, Hyeonae Im, Eunji Lee, Hyunjung Kang, Junhwa Seong, Yujin Park, Dohyun Kang, **Jeong Woo Han**, and Junsuk Rho
    Sci. Adv. **in revision** (2025). (IF=11.7)
41. Critical role of silanol defects in alkali-ZSM-5 on the selective dehydration of lactic acid to bio-acrylic acid
    Jichan Kim†, Hyunwoo Yook†, Jae-Soon Choi, Tae Hun Kang, Hoi-In Jeong, Dong-ha Lim, Raul F. Lobo, **Jeong Woo Han***, and Jungho Jae*
    J. Catal. **in revision** (2025). (IF=6.5)
42. Rh Cluster Catalysts with Enhanced Catalytic Activity: The 'Goldilocks Rh Size' for Olefin Hydroformylation
    Daewon Oh†, Miyeon Kim†, Hojeong Lee, Jong-Seong Bae, Beomgyun Jeong, Changhun Hur, **Jeong Woo Han***, and Kwangjin An*
    ACS Catal. **in revision** (2025). (IF=13.1)
43. Mechanistic Understanding of Nickel Catalyzed Urea Oxidation Reaction to Cyanate and Nitrite
    Kyu In Shim†, Jiseon Kim†, Miyeon Kim, Kangwoo Cho*, and **Jeong Woo Han***
    Exploration **in revision** (2025). (IF=22.5)
44. Understanding oxygen transfer on ceria with Pt single atoms for surface reaction

Yunji Choi†, Seokhyun Choung†, Jaebeom Han†, Jae-eon Hwang, Seungeun Kim, Jeong Young Park*, **Jeong Woo Han***, and Hyunjoo Lee*
Nat. Commun. **in press** (2025). (IF=15.7)

45. CatBench Framework for Benchmarking Machine Learning Interatomic Potentials in Adsorption Energy Predictions for Heterogeneous Catalysis
inuk Moon, Uchan Jeon, Seokhyun Choung, and **Jeong Woo Han***
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46. Strategic Design of Oxophilic Dopants for Active and Durable Alkaline Hydrogen Evolution Reaction under Seawater
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Ohhyun Kwon†, Tae Yong Kim†, Taewon Kim, Jihyeon Kang, Seohyeon Jang, Hojong Eom, Seyoung Choi, Junhyeop Shin, Jongkwon Park, Myeong-Lok Seol, **Jeong Woo Han**, Soomin Park, Hyun-Wook Lee, and Inho Nam
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Jihyeon Lee†, Byung Gwan Park†, Kiheon Sung†, Hyeongeon Lee, Jihun Kim, Eonu Nam, **Jeong Woo Han***, and Kwangjin An*
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Jaeyoung Lee†, Jiwoo Song†, Hyeonjung Jung†, Ilpyo Rho, Euiyoung Jung, **Jeong Woo Han***, and Taekyung Yu*
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146. Pseudocapacitive Behavior of Supercapattery Electrodes by Incorporating Schottky Junction for Ultrahigh Energy Density
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Yeongjun Yoon†, Hyo Min You†, Hyung Jun Kim, Matthew T. Curnan*, Kyeounghak Kim*, and **Jeong Woo Han***
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Zuyun He†, Jinwoo Hwang†, Zhiheng Gong, Mengzhen Zhou, Nian Zhang, Xiongwu Kang, **Jeong Woo Han***, and Yan Chen*
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164. Concurrent promotion of phase transition and bimetallic nanocatalyst exsolution in perovskite oxides driven by Pd doping to achieve highly active bifunctional fuel electrodes for reversible solid oxide electrochemical cells
Kyeong Joon Kim†, Chaesung Lim†, Kyung Taek Bae†, Jong Jun Lee, Mi Young Oh, Hyung Jun Kim, Hyunmin Kim, Guntae Kim, Tae Ho Shin*, **Jeong Woo Han***, and Kang Taek Lee*
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Yeji Choi†, Hyeonjung Jung†, Suji Kim, **Jeong Woo Han***, and Ki Bong Lee*
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Hyunmin Kim†, Chaesung Lim†, Ohhun Kwon†, Hu Young Jeong, Sihyuk Choi*, **Jeong Woo Han***, and Guntae Kim*
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354. Importance of Kinetics in Surface Alloying: A Comparison of the Diffusion Pathways of Pd and Ag Atoms on Cu(111)
Darin O. Bellisario, **Jeong Woo Han**, Heather L. Tierney, Ashleigh E. Baber, David S. Sholl, and E. Charles H. Sykes
J. Phys. Chem. C **113** (2009) 12863–12869. (IF=3.396)
355. Enantiospecific Adsorption of Amino Acids on Hydroxylated Quartz (0001)
Jeong Woo Han and David S. Sholl,
Langmuir **25** (2009) 10737–10745. (IF=4.097)
356. Step decoration of chiral metal surfaces
Jeong Woo Han, John R. Kitchin, and David S. Sholl
J. Chem. Phys. **130** (2009) 124710. (IF=3.149)
357. First principles calculations of methylamine and methanol adsorption on hydroxylated quartz (0001)
Jeong Woo Han, Joanna N. James, and David S. Sholl
Surf. Sci. **602** (2008) 2478–2485. (IF=1.731)
358. Characterization of the gel-spun tubular scaffold for cardiovascular tissue engineering
Eunna Chung, Sang-Heon Kim, Young Gun Ko, Jae-Hyun Kwon, **Jeong Woo Han**, In Su Park, Sung-Sik Han, and Soo Hyun Kim
Key Eng. Mater. **342–343** (2007) 321–324.
359. Optimization of scaffold for a successful hydrogel-seeding method for vascular tissue engineering
In Su Park, Sang-Heon Kim, **Jeong Woo Han**, Young Gun Ko, Eunna Chung, and Soo Hyun Kim
Key Eng. Mater. **342–343** (2007) 333–336.

INVITED CONFERENCE PRESENTATIONS

1. **Jeong Woo Han***, "Theory-Guided Design of Metal-Nitrogen Doped Carbon Single and Dual Atom Catalysts for Efficient Electrocatalytic Reactions", 4th International Symposium on Single-Atom Catalysis (ISSAC-4), *Keynote Presentation*, Singapore, August 4-7, 2025.
2. **Jeong Woo Han***, "Mechanistic Study for the Concurrent Process of Phase Transition and Exsolution and Its Application on Thermal Catalytic Reactions", 23rd International Conference on Solid State Ionics, *Keynote Presentation*, Boston, MA, USA, July 17-22, 2022.
3. **Jeong Woo Han***, "Phase Transition and Metal Exsolution in Perovskites to Enhance the Catalytic Activity for Hydrogen Production and Conversion", 2021 Virtual AIChE Annual Meeting, November 15-19, 2021.
4. **Jeong Woo Han***, "Tunable Nano-Distribution of Pt on TiO_2 Nanotubes by Atomic Compression Control for High-Efficient Oxygen Reduction Reaction", 6th ICAE, Jeju, Korea, November 9-12, 2021.
5. **Jeong Woo Han***, "Computational Analysis of Exsolution Phenomenon in Perovskite Oxide for Improved Catalytic Activity at the Anode of Solid Oxide Fuel Cell", 6th Asian SOFC, Jeju, Korea, November 7-10, 2021.
6. **Jeong Woo Han***, "Phase Transition and Metal Exsolution in Perovskites to Enhance the Catalytic Activity", IUMRS-ICA, Jeju, Korea, October 3-8, 2021.
7. **Jeong Woo Han***, "Rational design of ceria-based catalysts for enhanced catalytic activity", KIChE Spring Meeting, Busan, Korea, April 21-23, 2021.
8. **Jeong Woo Han***, "Rational Design of CeO_2-Based Catalysts for Enhanced Water-Gas Shift Reaction", 2020 Virtual AIChE Annual Meeting, November 16-20, 2020.
9. **Jeong Woo Han***, "Computational Design of Ex-solution Phenomena in Perovskite Oxide for Improved Catalytic Activity", KCS Fall Meeting, Suwon, Korea, October 19-21, 2020.
10. **Jeong Woo Han***, "Rational Design" of Metal Oxide Catalysts for Low-Temperature CO Oxidation", KIChE Fall Meeting, e-Conference, October 14-16, 2020.
11. **Jeong Woo Han***, "Computational Design of Perovskite-type Oxide Electrode to Achieve the Improved Catalytic Activity via Exsolution Phenomena", KECS Spring Meeting, Jeju, Korea, July 16-18, 2020.
12. **Jeong Woo Han***, "Computational Design of Perovskite-type Oxide Electrode to Achieve the Improved Catalytic Activity through Exsolution Phenomena", 44th International Conference and Exposition on Advanced Ceramics and Composites, Daytona Beach, FL, January 26-31, 2020.
13. **Jeong Woo Han***, "Computational Design of Catalysis", 2019 Inorganic Catalyst Symposium, Catalyst Development Center, LG Chem, Daejeon, Korea, February 26, 2019.
14. **Jeong Woo Han***, "Computational Materials Design for Developing High Performance Solid Oxide Fuel Cell Electrodes", 2019 Winter Symposium, Thermodynamics Division of KIChE, Daejeon, Korea, January 13-14, 2019.
15. **Jeong Woo Han***, "Computational Materials Design for Developing High Performance Solid Oxide Fuel Cell Electrodes", POSTECH-ShanghaiTech Symposium, Shanghai, China, January 15, 2019.

16. **Jeong Woo Han***, "Computational Materials Design for Hydrogen Economy", LOHC Workshop, Seoul, Korea, December 7, 2018.
17. **Jeong Woo Han***, "Computational Design of High Performance LOHC and the Catalysts for its Dehydrogenation", The 4th Korea-Japan Joint Symposium on Hydrogen in Materials, Buyeo, Korea, November 15-17, 2018.
18. **Jeong Woo Han***, "Mechanistic Study of Enhanced CO Oxidation Activity on Rare-Earth and Transition Metal Doped CeO_2", ENGE 2018, Jeju, Korea, November 11-14, 2018.
19. **Jeong Woo Han***, "Computational Design of Single-Atom Electrocatalysts for Hydrogen Evolution and Oxygen Reduction Reactions", 2018 International Symposium on Transport Phenomena and Applications, Yunlin, Taiwan, November 9-10, 2018.
20. **Jeong Woo Han***, "Rational Design of Rare-Earth and Transition Metal Doped CeO_2 Catalysts for Enhanced CO Oxidation Activity", 6th International Symposium on Processes Intensification, Taipei, Taiwan, November 7-8, 2018.
21. **Jeong Woo Han***, "Computational Design of Single-Atom Electrocatalysts for Hydrogen Evolution and Oxygen Reduction Reactions", 2018 AIChE Annual Meeting, Pittsburgh, PA, October 28-Novbember 2, 2018.
22. **Jeong Woo Han***, "Strain-driven ex-solution in perovskite oxide for improved catalytic activity", Korea Energy Transition Conference 2018, Seoul, Korea, October 4-5, 2018.
23. **Jeong Woo Han***, "Computational Materials Design for Developing High Performance Solid Oxide Fuel Cell Electrodes", POSTECH-Yonsei Symposium, Seoul, Korea, August 31, 2018.
24. **Jeong Woo Han***, "Computational Design of Single-Atom Electrocatalysts for Hydrogen Evolution and Oxygen Reduction Reactions", The 2nd Workshop on Functional Materials Science, Busan, Korea, October 22-23, 2018.
25. **Jeong Woo Han***, "Rational Design of Rare-Earth and Transition Metal Doped CeO_2 Catalysts for Enhanced CO Oxidation Activity", 4th Workshop on Supercomputing for Computational Bio/Nano/Materials Science, Korea Institute of Science and Technology Information, Daejeon, Korea, July 11, 2018.
26. **Jeong Woo Han***, "Computational Design of Catalysts for CO_2 Utilization", 2018 C1 Gas Refinery Symposium, Yangyang, Korea, July 4-6, 2018.
27. **Jeong Woo Han***, "Computational Materials Design for Developing High Performance Solid Oxide Fuel Cell Electrodes", 2017 NCU Ch&ME Workshop: Materials Chemistry and Physics, Taoyuan City, Taiwan, Novbember 9-11, 2017.
28. **Jeong Woo Han***, "Computational Materials Design for Developing High Performance Solid Oxide Fuel Cell Electrodes", 2017 AIChE Annual Meeting, Minneapolis, MN, October 29-Novbember 3, 2017.
29. **Jeong Woo Han***, "Effect of Metal Dopant on Ceria Catalysts for CO Oxidation", KIChE Fall Meeting, Daejeon, Korea, October 25-27, 2017.
30. **Jeong Woo Han***, "Computational Materials Design for Developing High Performance Solid Oxide Fuel Cell Electrodes", KIChE Fall Meeting, Daejeon, Korea, October 25-27, 2017.
31. **Jeong Woo Han***, "Computational Materials Design for Developing High Performance Solid Oxide Fuel Cell Electrodes", KPS Fall Meeting, Gyeongju, Korea, October 25-27, 2017.

32. **Jeong Woo Han***, "Computational Materials Design for Developing High Performance Solid Oxide Fuel Cell Electrodes", KCERS Fall Meeting, Seoul, Korea, October 11-13, 2017.
33. **Jeong Woo Han***, "Computational Materials Design for Developing High Performance Solid Oxide Fuel Cell Electrodes", SCEJ 49th Autumn Meeting, Nagoya, Japan, September 20-22, 2017.
34. **Jeong Woo Han***, "Computational Materials Design for Developing High Performance Solid Oxide Fuel Cell Electrodes", Nano Korea, Ilsan, Korea, July 12-14, 2017.
35. Hyunguk Kwon and **Jeong Woo Han***, "Computational Design of SOFC Cathode Materials with Enhanced Chemical Stability", 21st International Conference on Solid State Ionics, Padua, Italy, June 18-23, 2017.
36. Hyunguk Kwon and **Jeong Woo Han***, "Computational Design of High Performance Solid Oxide Fuel Cell Materials", 41st International Conference and Exposition on Advanced Ceramics and Composites, Daytona Beach, Florida, January 22-27, 2017.
37. **Jeong Woo Han***, "First-Principles based Rational Design of Catalyst", 2016 International Symposium on Transport Phenomena and Applications, Taoyuan, Taiwan, November 25-26, 2016.
38. **Jeong Woo Han*** and Kyeounghak Kim, "Origin of the Metallic Nanoparticle Exsolution on $ABB'O_3$ Perovskite", KCERS Fall Meeting, Seoul, Korea, November 23-25, 2016.
39. Kyeounghak Kim and **Jeong Woo Han***, "Density Functional Theory based Computational Screening of Metallic Nanoparticle Exsolution on $ABB'O_3$ Perovskites", The International Conference on Technological Advances of Thin Films & Surface Coatings, Singapore, July 12-15, 2016.
40. **Jeong Woo Han***, "First-Principles Calculations based Rational Design of SOFC Materials", KECS Spring Meeting, Gwangju, Korea, April 7-9, 2016.
41. **Jeong Woo Han***, Hyunguk Kwon, WooChul Jung, and Wonyoung Lee, "Factors Governing Cation Segregation at the Surfaces of Perovskite-type Materials: A Density Functional Theory Study", KCERS Fall Meeting, Incheon, Korea, November 4-6, 2015.
42. **Jeong Woo Han***, "First-Principle based Rational Design of Solid Oxide Fuel Cell Cathode Materials", IUMRS-ICAM, Jeju, Korea, October 25-29, 2015.
43. **Jeong Woo Han***, "Development of Chiral Surfaces and their Modifications for Enantiospecific Separation Processing", KCS Fall Meeting, Daegu, Korea, October 14-16, 2015.
44. Hyunguk Kwon, Wonyoung Lee, WooChul Jung, and **Jeong Woo Han***, "First-Principles based Design of High Performance Cathodes for Solid Oxide Fuel Cells", US-Korea Conference 2015, Atlanta, GA, July 29-August 1, 2015.
45. Jeonghyun Ko and **Jeong Woo Han***, "First-principles based Fast Screening of Binary Transition Metal Catalysts for CO_2 Conversion", US-Korea Conference 2015, Atlanta, GA, July 29-August 1, 2015.
46. Wonyoung Lee* and **Jeong Woo Han**, "Thermal Degradation at the Electrode Surfaces of Solid Oxide Fuel Cells", KCERS Spring Meeting, Pyeongchang, Korea, April 15-17, 2015.
47. **Jeong Woo Han***, Hyunguk Kwon, Wonyoung Lee, and WooChul Jung, "Density Functional Theory Study for Controlling Cation Segregation Behavior on SOFC Cathode Materials", KCERS Spring Meeting, Pyeongchang, Korea, April 15-17, 2015.

48. Jeonghyun Ko, Hyunguk Kwon, and **Jeong Woo Han***, "Fast Screening of Catalytic Activities on a Wide Range of Binary Transition Metal Alloys: Universality in Surface Mixing Rule of Adsorption Strength", MCARE 2015, Jeju, Korea, February 24-27, 2015.
49. Jeonghyun Ko, Hyunguk Kwon and **Jeong Woo Han***, "Surface Mixing Rule for Rapid Screening of Catalytic Activity on Binary Transition Metal Alloys", KCS Fall Meeting, Gwangju, Korea, October 15-17, 2014.
50. **Jeong Woo Han*** and Ho Seong Song, "Surface Modification and Functionalizing for Asymmetric Interfaces", UKC 2014, San Francisco, CA, August 6-9, 2014.
51. **Jeong Woo Han***, Kyeounghak Kim, Hyung Uk Kwon, and Wonyoung Lee, "Rational Design of Solid Oxide Fuel Cell Cathode Materials with Enhanced Oxygen Reduction Kinetics", ThinFilms 2014, Chongqing, China, July 15-18, 2014.
52. **Jeong Woo Han***, "Computational Design of SOFC Cathode Materials with Enhanced Oxygen Reduction Kinetics", 10th KIAS Electronic Structure Calculation Workshop, Seoul, Korea, June 19-20, 2014.
53. **Jeong Woo Han***, "Computational Design of Catalysis & Energy Materials", KSIEC Spring Meeting, Jeju, Korea, April 30-May 2, 2014.
54. Jeonghyun Ko, Hyejin Kang, Kyeounghak Kim, Byung-Kook Kim, and **Jeong Woo Han***, "First-principles based screening of binary transition metal catalysts for solid oxide electrolyzer cells", International Conference on Hydrogen Production, Fukuoka, Japan, February 2-5, 2014.
55. **Jeong Woo Han***, Jeonghyun Ko, Hyejin Kang, Kyeounghak Kim, and Byung-Kook Kim, "Density Functional Theory Based Screening of Binary Transition Metal Catalysts for Solid Oxide Electrolyzer Cells", The Korean Ceramic Society Fall Meeting, Jeju, Korea, October 16-18, 2013.
56. **Jeong Woo Han*** and Bilge Yildiz, "Oxygen Adsorption and Incorporation Mechanisms on $La_2CoO_{4+\delta}$ Surface", 18th International Conference on Solid State Ionics, Warszawa, Poland, July 3-8, 2011.

INVITED PRESENTATIONS

1. **Invited Talk**, Department of Chemical and Biomolecular Engineering, Seoul National University of Science and Technology, Seoul, Korea, January 2021.
2. **Invited Talk**, Korea Institute of Materials Science, Changwon, Korea, January 2020.
3. **Invited Talk**, Department of New Energy Engineering, Seoul National University of Science and Technology, Seoul, Korea, August 2019.
4. **Invited Talk**, Department of Materials Science & Engineering, Seoul National University, Seoul, Korea, July 2019.
5. **Invited Talk**, Department of Chemical Engineering, Inha University, Incheon, Korea, June 2019.
6. **Invited Talk**, National Institute of Environmental Research, Incheon, Korea, June 2019.
7. **Invited Talk**, Department of Energy & Chemical Engineering, Ulsan National Institute of Science and Technology (UNIST), Ulsan, Korea, June 2019.
8. **Invited Talk**, Catalysis Unit, SK Innovation, Daejeon, Korea, February 2019.
9. **Invited Talk**, Korea Institute of Energy Research (KIER), Daejeon, Korea, January 2019.
10. **Invited Talk**, Department of BIN Fusion Technology, Chonbuk National University, Jeonju, Korea, November 2018.

11. **Invited Talk**, Department of Chemical Engineering, Inha University, Incheon, Korea, May 2018.
12. **Tutorial**, Korean Institute of Chemical Engineers, Changwon, Korea, April 2018.
13. **Invited Talk**, The 1st DGIST Workshop on Energy Materials, Daegu Gyeongbuk Institute of Science and Technology (DGIST), Daegu, Korea, February 2018.
14. **Invited Talk**, The 5th Young Scientist Workshop, Department of Materials Science & Engineering, Seoul National University, Seoul, Korea, February 2018.
15. **Invited Talk**, Department of Chemical & Biomolecular Engineering, Sogang University, Seoul, Korea, January 2018.
16. **Invited Lecture**, Catalysis Lecture, Korean Institute of Chemical Engineers, Cheonan, Korea, January 2018.
17. **Invited Talk**, Biomass and Waste Energy Laboratory, Korea Institute of Energy Research (KIER), Jeju, Korea, December 2017.
18. **Invited Talk**, Department of Chemical Engineering and Material Science, Chung-Ang University, Seoul, Korea, December 2017.
19. **Department Seminar**, Department of Energy & Chemical Engineering, Ulsan National Institute of Science and Technology (UNIST), Ulsan, Korea, November 2017.
20. **Department Seminar**, Department of Chemical Engineering, Hanyang University, Seoul, Korea, October 2017.
21. **Department Seminar**, Department of Chemical Engineering, Pohang University of Science and Technology (POSTECH), Pohang, Korea, July 2017.
22. **Invited Talk**, 3rd Workshop on Supercomputing for Computational Bio/Nano/Materials Science, Korea Institute of Science and Technology Information, Daejeon. Korea, May 2017.
23. **Department Seminar**, School of Mechanical Engineering, Sungkyunkwan University, Suwon, Korea, March 2017.
24. **Invited Talk**, Hanwha Chemical R&D Center, Pangyo, Korea, January 2017.
25. **Department Seminar**, Department of Energy Systems Engineering, Daegu Gyeongbuk Institute of Science and Technology (DGIST), Daegu, Korea, October 2016.
26. **Invited Talk**, Next-generation Catalysis Center, Ulsan National Institute of Science and Technology (UNIST), Ulsan, Korea, August 2016.
27. **Invited Talk**, Center for Computational Science, Korea Institute of Science and Technology (KIST), Seoul, Korea, August 2016.
28. **Invited Talk**, Korea Research Institute of Chemical Technology (KRICT), Daejeon, Korea, July 2016.
29. **Invited Talk**, Department of Chemical & Biomolecular Engineering, Yonsei University, Seoul, Korea, June 2016.
30. **Invited Talk**, Department of Chemical & Biomolecular Engineering, Sogang University, Seoul, Korea, June 2016.
31. **Invited Talk**, Department of Organic Material Science & Engineering, Pusan National University, Busan, Korea, May 2016.
32. **Invited Talk**, School of Chemical & Biological Engineering, Seoul National University, Seoul, Korea, May 2016.
33. **Invited Talk**, Department of Chemical & Biological Engineering, Korea University, Seoul Korea, April 2016.
34. **Invited Talk**, Functional Crystallization Center Workshop, Jeongseon, Korea, February 2016.
35. **Invited Lecture**, Catalysis Lecture, Korean Institute of Chemical Engineers, Cheonan, Korea, January 2016.

36. **Invited Talk**, Korea Institute of Materials Science, Changwon, Korea, January 2016.
37. **Department Seminar**, Department of Chemical Engineering, Ajou University, Suwon, Korea, January 2016.
38. **Department Seminar**, Department of Chemical Engineering, Inha University, Incheon, Korea, December 2015.
39. **Invited Talk**, School of Chemical & Biological Engineering, Seoul National University, Seoul, Korea, August 2015.
40. **Invited Talk**, Jeju Global Research Center, Korea Institute of Energy Research (KIER), Jeju, Korea, July 2015.
41. **Invited Talk**, Hydrogen and Fuel Cell Center for Industry, Academy, and Laboratories, Korea Institute of Energy Research (KIER), Buan, Korea, June 2015.
42. **Department Seminar**, Department of Environment and Energy Engineering, Myongji University, Yongin, Korea, January 2015.
43. **Invited Talk**, Next Generation HPC System, Seoul, Korea, September 2014.
44. **Department Seminar**, Department of Environment and Energy Engineering, Myongji University, Yongin, Korea, August 2014.
45. **Invited Talk**, School of Energy and Chemical Engineering, Ulsan National Institute of Science and Technology (UNIST), Ulsan, Korea, June 2014.
46. **Department Seminar**, Department of Materials Science & Engineering, Seoul National University, Seoul, Korea, May 2014.
47. **Invited Talk**, Department of Naval Architecture & Ocean Engineering, Seoul National University, Seoul, Korea, April 2014.
48. **Invited Talk**, Korea Institute of Energy Research (KIER), Daejeon, Korea, February 2014.
49. **Invited Talk**, Hydrogen & Fuel Cell Academy, Hanyang University, Seoul, Korea, February 2014.
50. **Invited Talk**, Department of Chemical Engineering, Pohang University of Science and Technology (POSTECH), Pohang, Korea, January 2014.
51. **Invited Lecture**, Catalysis Lecture, Korean Institute of Chemical Engineers, Cheonan, Korea, January 2014.
52. **Invited Talk**, Korea Research Institute of Chemical Technology (KRICT), Daejeon, Korea, July 2013.
53. **Invited Talk**, Graduate School of Energy, Environment, Water and Sustainability, Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Korea, July 2013.
54. **Invited Lecture**, School of Chemical & Biological Engineering, Seoul National University, Seoul, Korea, April 2013.
55. **International Micro Solid Oxide Fuel Cell Workshop**, Korea University, Seoul, Korea, October 2012.
56. **Invited Talk**, Department of Chemical Engineering, University of Seoul, Seoul, Korea, June 2012.
57. **Invited Talk**, School of Chemical Engineering, Sungkyunkwan University, Suwon, Korea, June 2012.
58. **Invited Talk**, Department of Chemical Engineering, Kyung Hee University, Yongin, Korea, June 2012.
59. **Department Seminar**, Department of Chemical & Biomolecular Engineering, Sogang University, Seoul, Korea, April 2012.
60. **Invited Talk**, BK21 Symposium, Department of Chemical & Biomolecular Engineering, Yonsei University, Seoul, Korea, September 2011.

61. **Invited Talk**, Young Scientist Workshop, Department of Materials Science & Engineering, Seoul National University, Seoul, Korea, September 2011.
62. **Invited Talk**, Young Investigator Seminar Series, Department of Chemical & Biomolecular Engineering, Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Korea, July 2011.
63. **Invited Talk**, Biomaterials Research Center, Korea Institute of Science and Technology (KIST), Seoul, Korea, December 2009.

PROJECTS / GRANTS (*Currently, > ₩1,900,000,000*)

1. **Jeong Woo Han** (PI), Hyundai Motor Company, Green & Eco-technology Team. (Ongoing)
2. **Jeong Woo Han** (PI), LG Chem, Pioneering Research Group. Future Technology. (Ongoing)
3. **Jeong Woo Han** (PI), LG Chem, Catalyst R&D. Catalyst Division. Petrochemicals company. (Ongoing)
4. **Jeong Woo Han** (PI), Hanwha Solutions, Chemical Division. Research & Development Institute. (Ongoing)
5. **Jeong Woo Han** (PI), Korea Shipbuilding & Offshore Engineering, Future Ship Research(or Technology) Laboratory. (Ongoing)
6. **Jeong Woo Han** (PI), Korea Shipbuilding & Offshore Engineering, Energy System Research Institute. (Ongoing)
7. **Jeong Woo Han** (PI), "Rational design of atomically dispersed metal alloy catalysts nano-socketed into the supports for highly efficient hydrogen production", Mid-Career Grant (Type 2) from the National Research Foundation of Korea. (Ongoing)
8. **Jeong Woo Han** (PI), "Computational Design of LOHC Materials and the Catalysts for Hydrogenation/Dehydrogenation", Hydrogen Energy Innovation Technology Development from the National Research Foundation of Korea. (Ongoing)
9. **Jeong Woo Han** (Co-PI), "Atomic-level surface control of complex oxides for electrochemical energy conversion", Nano·Material Technology Development Program from the National Research Foundation of Korea. (Ongoing)
10. **Jeong Woo Han** (Co-PI), "Materials discovery for single atom catalysts facilitating an independent control of adsorption strength of reaction intermediates", Future Material Discovery from the National Research Foundation of Korea. (Ongoing)
11. **Jeong Woo Han** (Co-PI), " Super Ultra Low Energy & Emission Vehicle Center", Engineering Research Center (ERC) from the National Research Foundation of Korea. (Ongoing)
12. **Jeong Woo Han** (Co-PI), "In-situ growth of self-assembled metal nano-catalysts using spontaneous metal/oxide phase separation", Samsung Research Funding Center for Future Technology. (Ongoing)
13. **Jeong Woo Han** (PI, 위탁), "Computational design of highly selective catalyst for PET to BTX production via hydrogenolysis and decarboxylation ", National Research Foundation of Korea. (Ongoing)
14. **Jeong Woo Han** (PI, 위탁), "Development of nano catalyst materials and cells for low-temperature direct ammonia SOFC", Nano·Material Technology Development Program from the National Research Foundation of Korea. (Ongoing)

15. **Jeong Woo Han** (PI, 위탁), "Establishment of computational chemistry model for optimal catalyst design", Carbon to X Technology Development from the National Research Foundation of Korea. (Ongoing)
16. **Jeong Woo Han** (PI, 위탁), "Computational Design of highly selective catalyst for ketonization and highly stable catalyst for aldol condensation", National Research Foundation of Korea. (Ongoing)
17. **Jeong Woo Han** (PI, 위탁), "Computational design of high performance catalysts for methane direct conversion", C1 Gas Refinery R&D Center from the National Research Foundation of Korea. (Ongoing)
18. **Jeong Woo Han** (PI, 위탁), "Computational study of high performance catalysts for methane-to-aromatics", C1 Gas Refinery R&D Center from the National Research Foundation of Korea. (Ongoing)
19. **Jeong Woo Han** (PI, 위탁), "Computational design of composite catalysts for SOFCs", Nano·Material Technology Development Program from the National Research Foundation of Korea. (Completed)
20. **Jeong Woo Han** (Co-PI), "Development of Water-Stable Nano-Phosphor with Ultra-Narrow Full-Width at Half-Maximum for Display Application", Samsung Research Funding Center for Future Technology. (Completed)
21. **Jeong Woo Han** (PI), "Rational Design of Enhanced Metal Catalyst Supported on Metal Oxide for Water-Gas Shift Reaction: from Computational Chemistry to Experimental Validation", Mid-Career Grant from the National Research Foundation of Korea. (Completed)
22. **Jeong Woo Han** (Co-PI), "Development of single-atom electrocatalysts for H_2 production and utilization", Samsung Research Funding Center for Future Technology. (Completed)
23. **Jeong Woo Han** (PI, 위탁), "Computational design of catalysis/support with high performance and stability for high temperature co-electrolysis", KIST Institution Program. (Completed)
24. **Jeong Woo Han** (Co-PI), "Development of liquid organic hydrogen carriers as a renewable storage and transport using modified heat-transfer fluids", Korea Institute of Energy Technology Evaluation and Planning. (Completed)
25. **Jeong Woo Han** (Co-PI), "Research on technology and commercialization of metal-supported solid oxide fuel cell for utilization of hydrocarbon fuels", Korea Institute of Energy Technology Evaluation and Planning. (Completed)
26. **Jeong Woo Han** (Co-PI), "The multiscale technology for long-term stabilized SOFC", Global Frontier Center for Multiscale Energy System from the National Research Foundation of Korea. (Completed)
27. **Jeong Woo Han** (PI), "Development of semiconductor materials through kinetic Monte Carlo based multiscale simulations", SK Hynix. (Completed)
28. **Jeong Woo Han** (PI), "Rational design of key materials for high performance solid oxide and proton exchange membrane fuel cells via computational materials science", Global Frontier Center for Multiscale Energy System from the National Research Foundation of Korea. (Completed)

29. **Jeong Woo Han** (PI), "Computational rational design of catalysis for eco-friendly energy sources: from co-electrolysis to synthetic hydrocarbon fuel", New Faculty Grant from the National Research Foundation of Korea. (Completed)
30. **Jeong Woo Han** (PI, 위탁), "Computational rational design of catalysis with high performance, selectivity, and stability for high temperature co-electrolysis", KIST Institution Program. (Completed)

Ph.D. GRADUATE STUDENTS SUPERVISED

1. **Kyeounghak Kim** (Graduated 2/2021) [Assistant Professor at Hanyang University]
2. **Rui Huang** (Graduated 8/2022) [Senior Researcher at LG Chem]
3. **Ara Cho** (Graduated 2/2023) [Postdoc at Stanford University]
4. **Kyung-Jong Noh** (Graduated 8/2023) [Project Manager at Umicore]
5. **Hyung Jun Kim** (Graduated 8/2023) [Senior Researcher at Hyundai Motor Company]
6. **Dongjae Shin** (Graduated 8/2023) [Postdoc at Stanford University]
7. **Hyeonjung Jung** (Graduated 8/2023) [Postdoc at Stanford University]
8. **Myeong Gon Jang** (Graduated 8/2023) [Senior Researcher at Samsung Electronics]
9. **Chaesung Lim** (Graduated 8/2023) [Senior Researcher at Samsung Display]
10. **Byoung Joon Park** (Graduated 8/2023) [Postdoc the University of Bayreuth]
11. **Hyunwoo Yook** (Graduated 8/2023) [Postdoc at Seoul National University]
12. **Jinwoo Hwang** (Graduated 2/2024) [Postdoc at University of Wisconsin-Madison]
13. **Kyu In Shim** (Graduated 2/2024) [Senior Researcher at HD Korea Shipbuilding & Offshore Engineering]
14. **Seokhyun Choung** (Graduated 2/2024) [Postdoc at Seoul National University]
15. **Yejung Choi** (Graduated 8/2025) [Senior Researcher at Samsung Electronics]

M.S. GRADUATE STUDENTS SUPERVISED

1. **Ho Seong Song** (Graduated 8/2016) [Account Manager at Merck]
2. **Jeonghyun Ko** (Graduated 8/2016) [Senior Researcher at SK On]
3. **Hyunguk Kwon** (Graduated 8/2017) [Assistant Professor at Seoul National Univ. of Science and Technology]
4. **Bohyun Hwang** (Graduated 2/2017)
5. **Jae Yul Lim** (Graduated 2/2019)
6. **Jihyeon Song** (Graduated 8/2023) [Researcher at HD Korea Shipbuilding & Offshore Engineering]
7. **Heejae Yang** (Graduated 2/2024) [Researcher at Samsung SDI]

RESEARCH PROFESSORS SUPERVISED

1. **Sandip Maiti** [Research Professor at POSTECH]
2. **Mathew T. Curnan** [Postdoc at KENTECH]

POSTDOCS SUPERVISED

1. **Suman K. Sahoo** [Assistant Professor at Jain University, India]
2. **Ismail Ibrahim** [Assistant Professor at Helwan University, Egypt]
3. **Ying Wang** [Research Assistant Professor at Hong Kong Polytechnic Univ., Hong Kong]

4. **Kakali Maiti** [Research Scientist at CSIR-Institute of Minerals and Materials Technology, India]
5. **Tae Yong Kim** [Assistant Professor at Korea Univ. of Technology & Education]

PROFESSIONAL ACTIVITIES

1. Made a number of the figures for the book*, "Density Functional Theory: A Practical Introduction"*, David S. Sholl & Janice A. Steckel, John Wiley & Sons, Inc., Hoboken, NJ.
2. Journal Reviewer:
   - *Nature Synthesis*; *Nature Catalysis*; *Nature Chemistry*; *Nature Communications*; *Science Advances; Journal of the American Chemical Society*; *Angewandte Chemie-International Edition; Advanced Materials; Chem; ACS Catalysis; Journal of Catalysis*; *Journal of Materials Chemistry A; Chemical Science; Chemical Engineering Journal*; *Langmuir*; *Journal of Physical Chemistry C*; *Journal of Physical Chemistry Letters*; *Journal of Chemical Physics*; *Physical Chemistry Chemical Physics*; *Surface Science*; *Applied Surface Science*; *Journal of Molecular Catalysis A: Chemical*; *Chemical Physics Letters*; *Korean Journal of Chemical Engineering*; *Journal of Electroceramics*; *Materials Science and Engineering B*; *Journal of Materials Science*; *Journal of Industrial and Engineering Chemistry*; *Journal of Fuel Cell Science and Technology*; *Journal of Korean Ceramic Society*; *Composites Science and Technology*; *Thin Solid Films*; *Journal of Alloys and Compounds*; *Applied Catalysis A: General; Applied Catalysis B: Environmental*; *Journal of Power Sources*; *Journal of CO_2 Utilization*; *ChemPhysChem*; *Technology*; *Surface and Coatings Technology; Ionics; Solid State Ionics; Catalysis Today*

PROFESSIONAL MEMBERSHIPS

1. Korean Institute of Chemical Engineers
2. Korean Society of Industrial and Engineering Chemistry
3. Korean Ceramic Society
4. Korean Chemical Society
5. Korean Electrochemical Society
6. American Institute of Chemical Engineers
7. Materials Research Society
8. European Materials Research Society
9. American Ceramic Society